

# Pharma Agentic AI – Analysis Report

**Query:** Perform a complete end-to-end opportunity assessment for the molecule MOLB in the Japan Asthma market including market sizing trends, import–export dependencies, patent barriers, clinical trial landscape, internal strategic insights, and alignment with the latest GINA guidelines. Identify unmet needs, repurposing opportunities, whitespace areas, key risks, and recommended product concepts for future portfolio expansion.

**Generated on:** 07 Dec 2025, 10:25 PM

**Report ID:** 48f5a2cf-2cfe-4697-b513-0b3a22723fee

## Executive Summary

Market & growth view: Total market size: 1603.2 Mn USD with average 5Y CAGR 11.2%. Records: 5. Supply and trade dynamics: Total imports: 9352.3 tonnes, total exports: 9995.7 tonnes across 188 records. IP barriers and freedom-to-operate: Found 48 relevant patents. Earliest expiry year: 2020. Clinical development and pipeline: Found 25 clinical trials matching the criteria. Internal strategy and unmet needs: Asthma market in Japan faces significant unmet needs, including steroid-free options, safer formulations for elderly, alternative therapies for steroid intolerance, low-dose options for pediatric cases, and inhaler affordability issues. MOLY presents opportunities for repurposing and addressing these needs. Guidelines and real-world evidence: The Global Initiative for Asthma (GINA) 2024 emphasizes the preference for inhaled corticosteroids (ICS)-based inhalers, while real-world evidence indicates low adherence and suboptimal inhaler use in various regions. Combination therapy and early risk stratification are crucial for moderate to severe asthma. Moreover, non-pharmacological interventions such as smoking cessation and vaccination are essential in asthma management. Regular follow-up and objective monitoring of disease control are recommended.

## IQVIA Insights Agent

**Summary:** Total market size: 1603.2 Mn USD with average 5Y CAGR 11.2%. Records: 5.

**Details:**

**Table:**

- **molecule:** MOLY **indication:** Asthma **country:** Japan **year:** 2021 **sales\_usd\_mn:** 445.41 **cagr\_5y:** 10.89 **competitors:** 12 **patient\_burden\_index:** 0.39
- **molecule:** MOLY **indication:** Asthma **country:** Japan **year:** 2021 **sales\_usd\_mn:** 275.13 **cagr\_5y:** 12.79 **competitors:** 11 **patient\_burden\_index:** 0.93
- **molecule:** MOLY **indication:** Asthma **country:** Japan **year:** 2023 **sales\_usd\_mn:** 223.51 **cagr\_5y:** 13.88 **competitors:** 6 **patient\_burden\_index:** 0.34
- **molecule:** MOLY **indication:** Asthma **country:** Japan **year:** 2024 **sales\_usd\_mn:** 104.8 **cagr\_5y:** 14.91 **competitors:** 4 **patient\_burden\_index:** 0.41
- **molecule:** MOLY **indication:** Asthma **country:** Japan **year:** 2022 **sales\_usd\_mn:** 554.33 **cagr\_5y:** 3.74 **competitors:** 7 **patient\_burden\_index:** 0.84

## EXIM Trade Agent

**Summary:** Total imports: 9352.3 tonnes, total exports: 9995.7 tonnes across 188 records.

### Details:

#### Table:

- **molecule:** MOLY **formulation:** API **country:** Japan **year:** 2021 **import\_tonnes:** 66.38 **export\_tonnes:** 17.77
- **molecule:** MOLY **formulation:** Capsule **country:** Japan **year:** 2020 **import\_tonnes:** 28.59 **export\_tonnes:** 90.19
- **molecule:** MOLY **formulation:** Inhaler **country:** Japan **year:** 2023 **import\_tonnes:** 30.66 **export\_tonnes:** 94.64

- **molecule:** MOLY **formulation:** Tablet **country:** Japan **year:** 2023 **import\_tonnes:** 77.86 **export\_tonnes:** 68.03
- **molecule:** MOLY **formulation:** Inhaler **country:** Japan **year:** 2024 **import\_tonnes:** 24.29 **export\_tonnes:** 6.75
- ... (5 more records)

## Patent Landscape Agent

**Summary:** Found 48 relevant patents. Earliest expiry year: 2020.

### Details:

#### Patents:

- **patent\_id:** US3023842 **molecule:** MOLY **indication:** Asthma **assignee:** Sunrise Therapeutics **filing\_year:** 2015 **expiry\_year:** 2035 **status:** Granted **fto\_flag:** High
- **patent\_id:** US8690566 **molecule:** MOLY **indication:** Asthma **assignee:** NovaCure Biotech **filing\_year:** 2006 **expiry\_year:** 2026 **status:** Pending **fto\_flag:** Medium
- **patent\_id:** US1634839 **molecule:** MOLY **indication:** Asthma **assignee:** CardioWell Pharma **filing\_year:** 2002 **expiry\_year:** 2022 **status:** Pending **fto\_flag:** Low
- **patent\_id:** US8433777 **molecule:** MOLY **indication:** Asthma **assignee:** GenericPharma Ltd **filing\_year:** 2012 **expiry\_year:** 2032 **status:** Granted **fto\_flag:** High
- **patent\_id:** US4523953 **molecule:** MOLY **indication:** Asthma **assignee:** Sunrise Therapeutics **filing\_year:** 2022 **expiry\_year:** 2042 **status:** Granted **fto\_flag:** High
- ... (5 more records)

## Clinical Trials Agent

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**Summary:** Found 25 clinical trials matching the criteria.

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**Details:**

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**Table:**

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- **trial\_id:** CT-00031   **molecule:** MOLY   **indication:** Asthma   **phase:** Phase II   **status:** Suspended   **sponsor:** NovaCure   **country:** Japan
  - **trial\_id:** CT-00713   **molecule:** MOLY   **indication:** Asthma   **phase:** Phase II   **status:** Suspended   **sponsor:** GlobalBio   **country:** Japan
  - **trial\_id:** CT-00801   **molecule:** MOLY   **indication:** Asthma   **phase:** Phase II   **status:** Suspended   **sponsor:** NeoHealth   **country:** Japan
  - **trial\_id:** CT-01009   **molecule:** MOLY   **indication:** Asthma   **phase:** Phase I   **status:** Recruiting   **sponsor:** PrimeCare   **country:** Japan
  - **trial\_id:** CT-02829   **molecule:** MOLY   **indication:** Asthma   **phase:** Phase III   **status:** Active   **sponsor:** ABC Pharma   **country:** Japan
  - ... (5 more records)
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## Internal Knowledge Agent

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**Summary:** Asthma market in Japan faces significant unmet needs, including steroid-free options, safer formulations for elderly, alternative therapies for steroid intolerance, low-dose options for

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pediatric cases, and inhaler affordability issues. MOLY presents opportunities for repurposing and addressing these needs.

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**Details:**

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**Documents:**

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- asthma\_strategy\_2024.pdf
  - respiratory\_field\_insights.pdf
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**Key Opportunities:**

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- MOLY once-daily inhaler for improved adherence and better patient experience
  - Steroid-free formulation for sensitive patients and elderly population
  - Low-dose inhaler for pediatric asthma cases and vulnerable patients
  - Inhaler affordability solutions for Tier-2/3 cities and rural areas
  - Alternative therapies for steroid intolerance and high-risk patients
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**Key Risks:**

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- Patent barriers and intellectual property challenges
  - Evidence gaps in clinical trial data for MOLY in asthma
  - Operational constraints due to regulatory complexities and approval processes
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- Competitive market dynamics and established brand presence in Japan
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- Safety and efficacy concerns related to MOLY in specific patient populations
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## Web Intelligence Agent

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**Summary:** The Global Initiative for Asthma (GINA) 2024 emphasizes the preference for inhaled corticosteroids (ICS)-based inhalers, while real-world evidence indicates low adherence and suboptimal inhaler use in various regions. Combination therapy and early risk stratification are crucial for moderate to severe asthma. Moreover, non-pharmacological interventions such as smoking cessation and vaccination are essential in asthma management. Regular follow-up and objective monitoring of disease control are recommended.

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### Details:

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#### Links:

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- **title:** Asthma guideline (synthetic) **url:** <https://example-guideline.com>
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- **title:** Asthma RWE summary **url:** <https://example-rwe.com>
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#### Key Updates:

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- GINA 2024 prefers ICS-based inhalers over oral corticosteroids.
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- Combination therapy is advised for moderate to severe asthma when monotherapy is insufficient.
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